Exercises Brief

*(*) The below exercise has been developed with **JDK 21**and **JavaFX 21.0.6** with Visual Studio Code for MacOS Sonoma 14.7.1.*

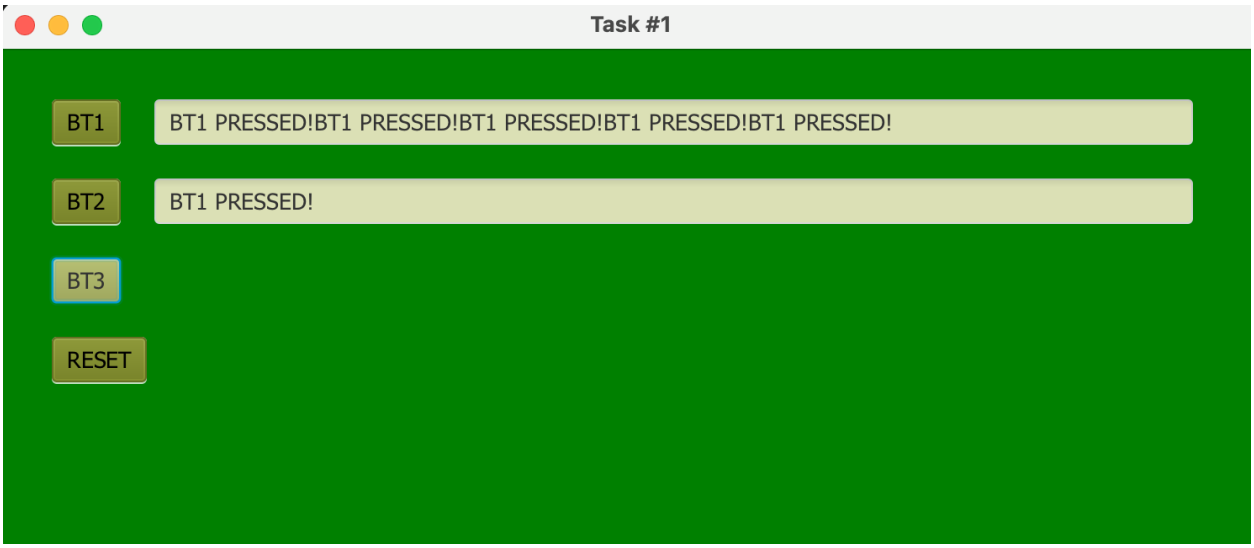
Programming II is one of the courses I loved the most because it gave me creativity and I can show off my exited ideas. I can raise my imagination and the code not just barely commands lines but becomes more interesting.

Below is briefly of previous exercises. To save time, I captured some highlight functions, but you can review my full programs directly anytime in the attachment and hopefully, you will give me some questions regarding the idea, functionalities, logic, or any issue may arise.

Exercise 1 to 5: Simple programs with console log

Exercise 6:

Task #1:



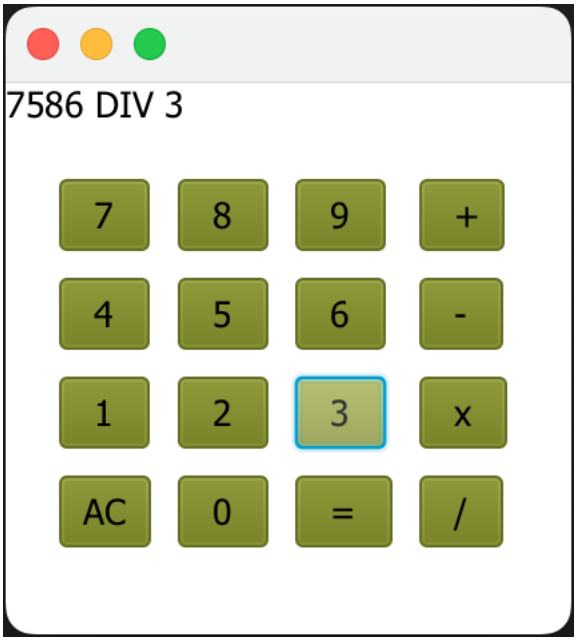
BT1: Inform the button pressed

BT2: Clone text from the first textbox to the second one

BT3: Change the form's background

Reset: Rollback all values of form's components to the original values

Task #2:



I created a calculator and implemented some simple operators. The use can use mouse to click on the buttons or using keyboard (enter to calculate)

Task #3: I created a matching game.

The user can:

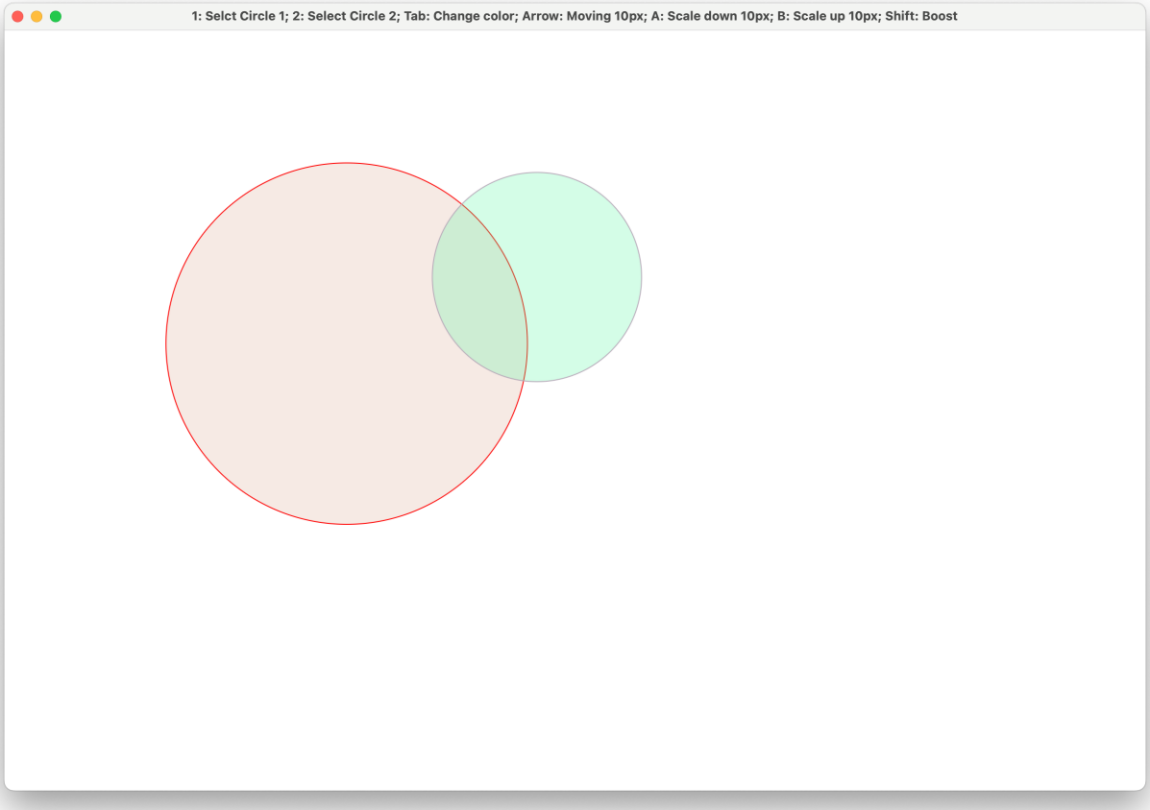
- Adjust the board size, and number of pairs can change accordingly

- Increase or decrease the image size by using the spinner

- Shuffle Images: create a new game and re-scatter pairs on the board
- No Shuffle Images: the pairs are arranged close together for easy testing

- Check the result with the notification at the bottom
- Tracking mouse position by swap the image color when the mouse hover on the images

Task #4:

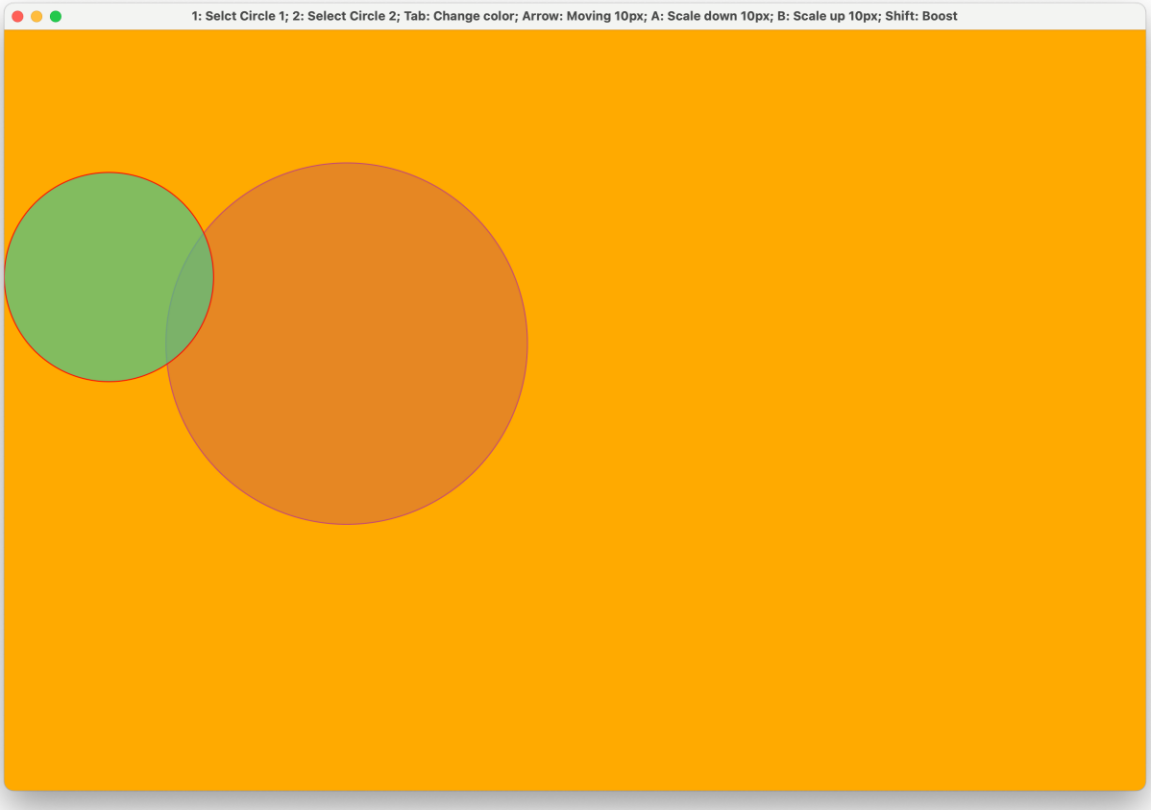


Press either key ‘1’ or ‘2’ to select the #1 or #2 circle then perform one of those actions:

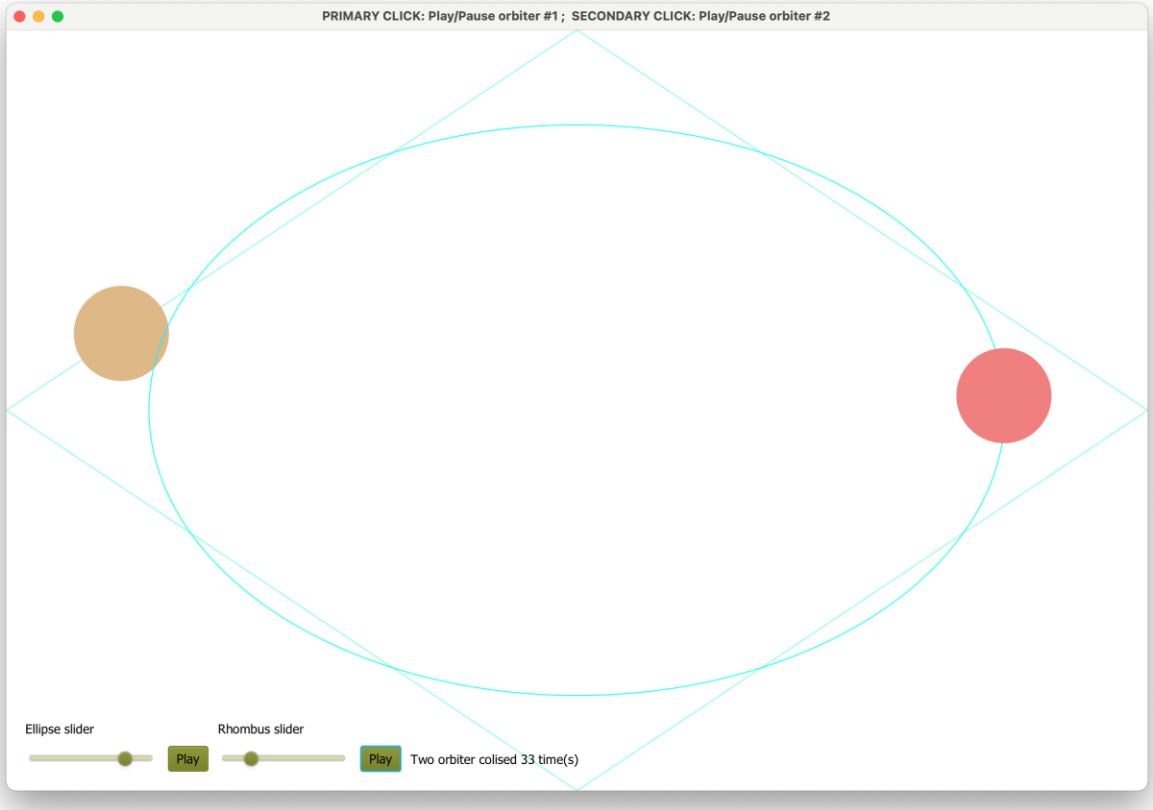
- When a circle is selected, its outline change to RED, deselected then its outline change to a random color
- Tab: Change fill color to a random color
- Arrow key: moving the selecting circle
- A: Increase the size of selecting circle
- B: Decrease the size of selecting circle
- Shift: Boost moving or scaling speed against the selecting circle

Global affects:

- If the next moving or scaling up will make the circle border out of the box, the form’s background will change color to ORANGE
- Circle color has Alpha parameter that allow overlay merge with other circle filling color

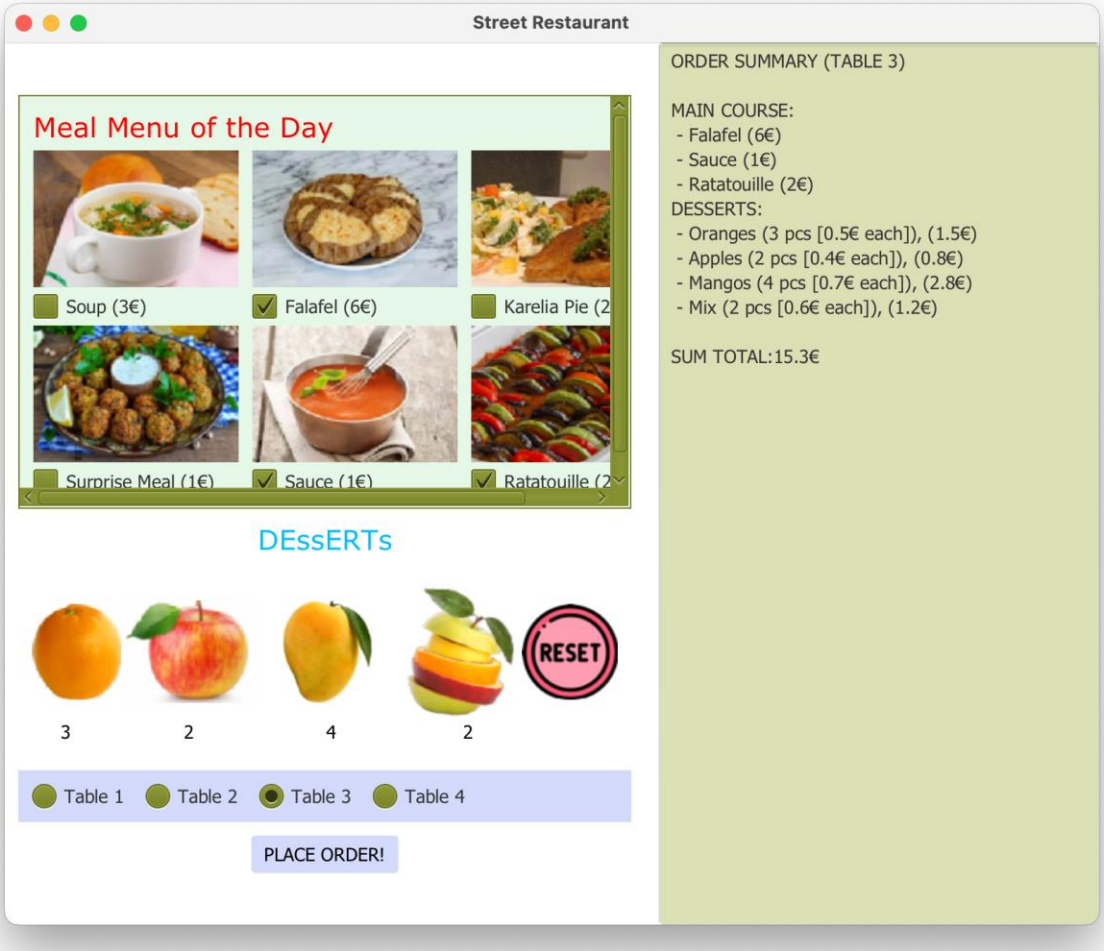


Task #5: Handle shape transition follows a path

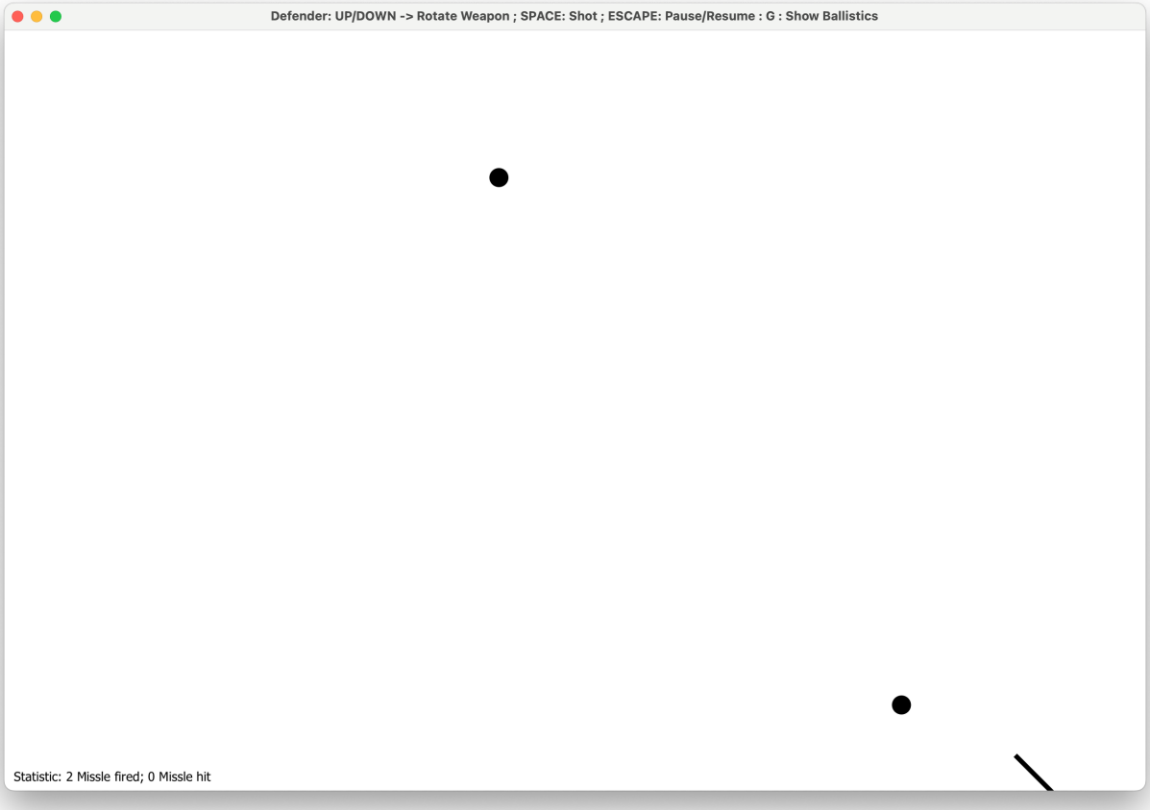


- A circle follows an Ellipse, and a circle follows Rhombus.
- Each circle has its slider to adjust the transition speed.
- When two circles touch together at their borders, the counter will increase by one. This operator can perform by using BooleanBinding, check if the distance of two circle centers falls into 2x circle diameter.

Task #6: Implement closed to the exercise requirements and there is no creative (because I do not like to break up the original coding style)

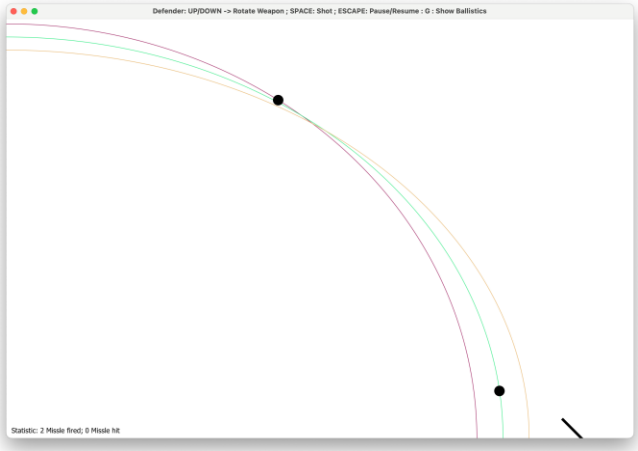


Task #7: I create a game like an Anti-missiles artillery system

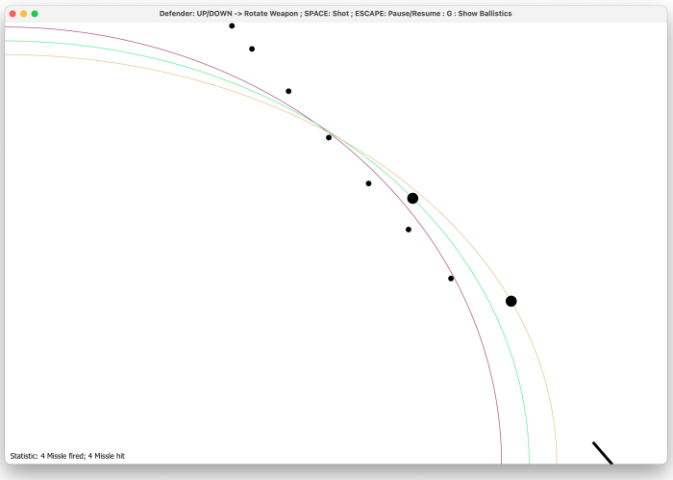


Basic rules:

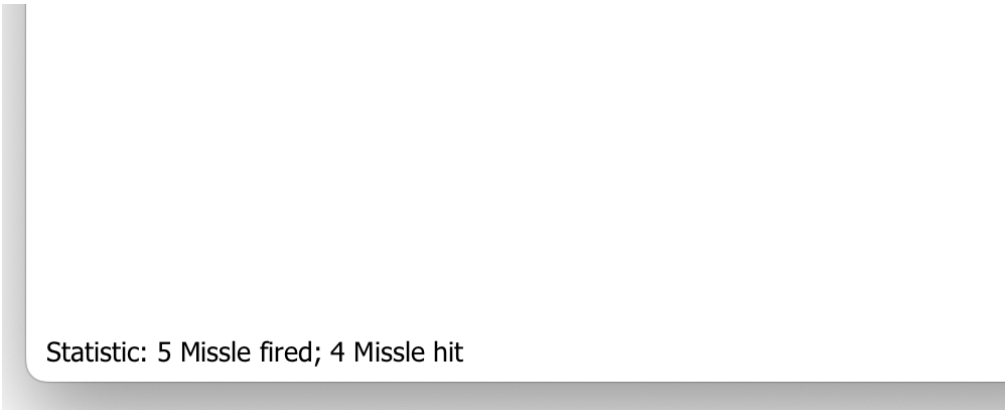
- Missiles fly on curved trajectory which can be shown by pressing the ‘G’ key. There are three ballistics which are used as transition paths for three missiles.



- The defend weapon is set on the ground, at the right corner.
- The weapon barrel can raise or lower aim on oncoming missiles from above.
- Press SPACE to trigger the weapon to fire bullets.

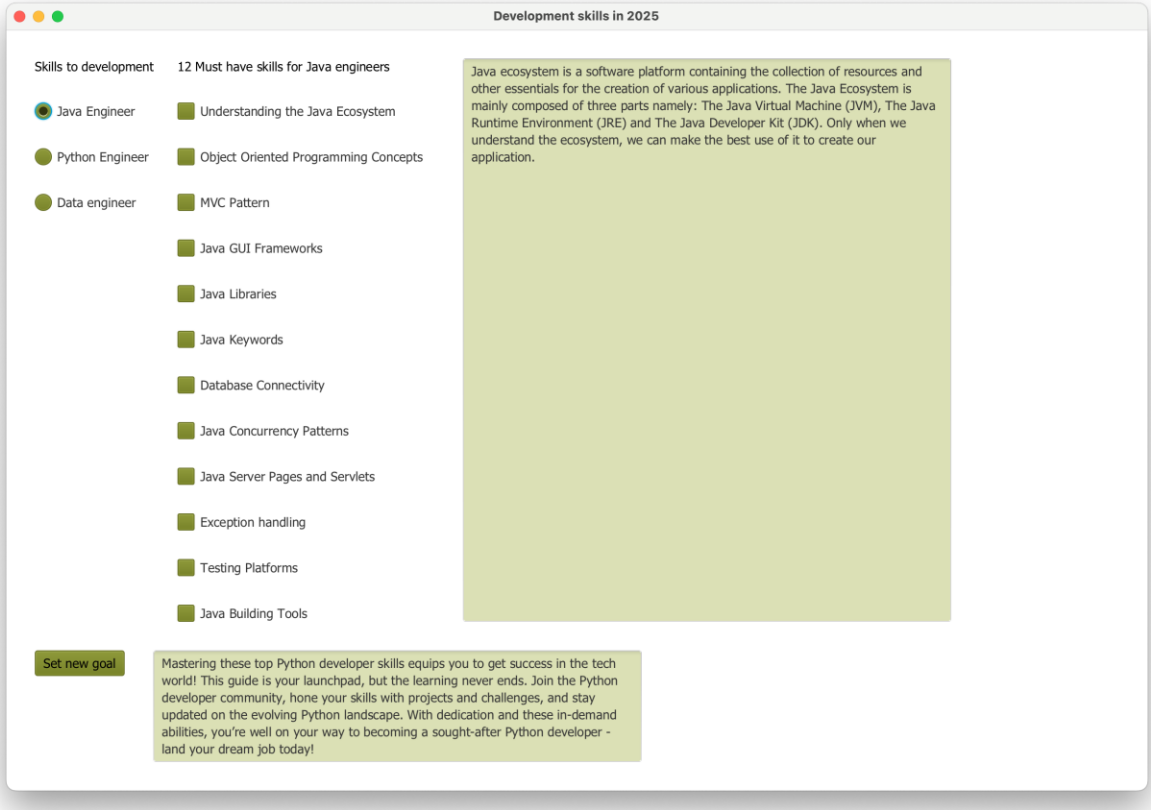


- Press ESCAPE to pause the game
- The game count number of missile fired and how many bullets fire AND hit the missile (by using BooleanBinding, similarity mechanical with Task #5)



Exercise 7:

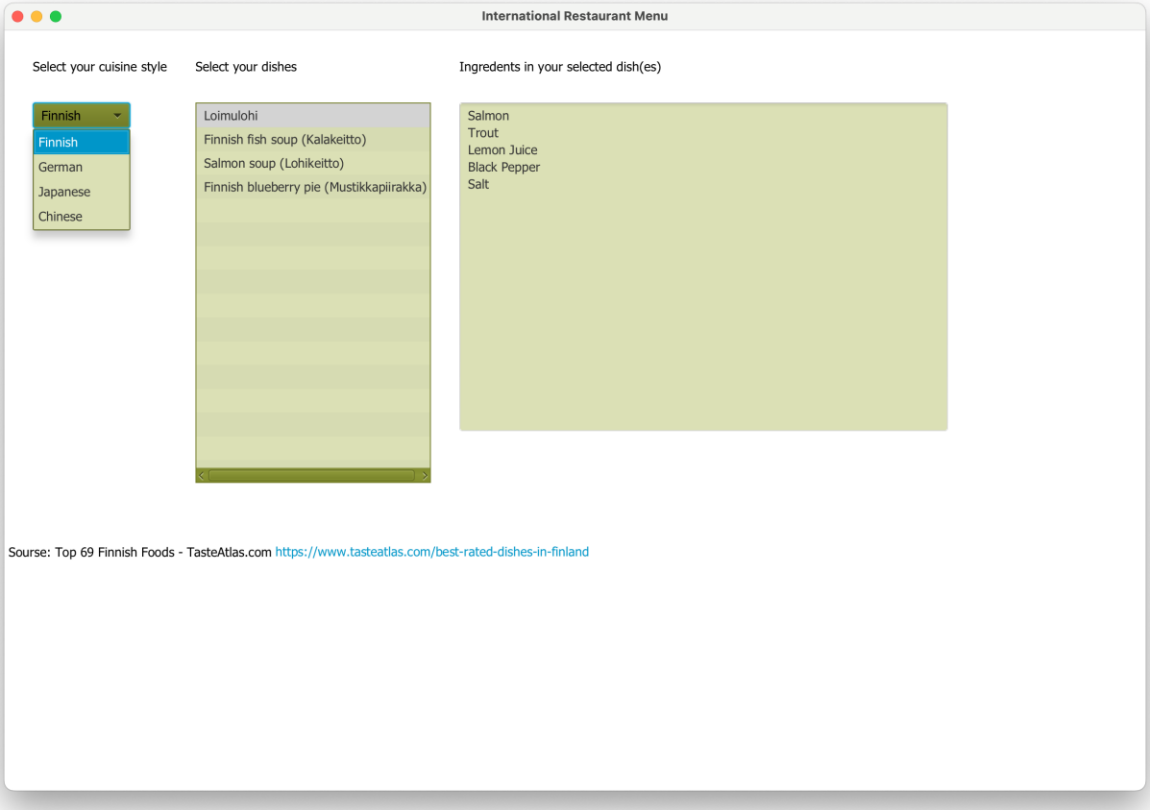
Task #1: Set goal for development skills



Basic rules:

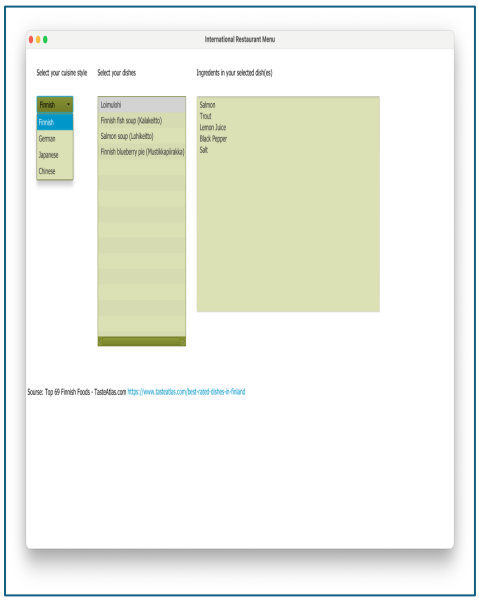
- Users select one of development competency: Java, Python or Data
- Each competency has
    - Its description in lower text area
    - A list of sub-skills needs to be mastered which each description in the right text area when hovering on it. The list can multi select
- Set new goal: the program will collect all selected sub-skill and set a new target for the coming year

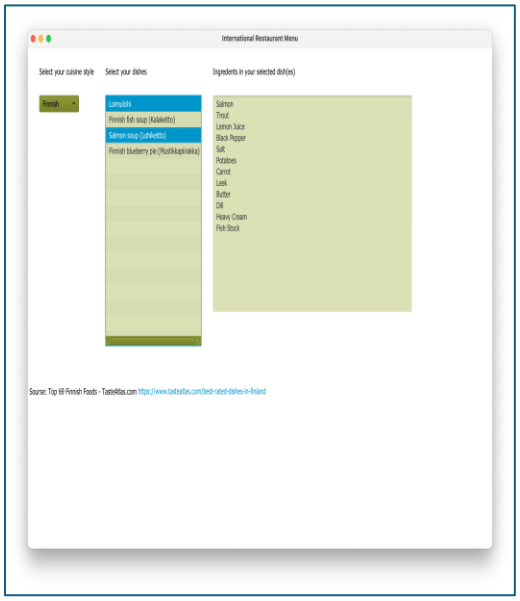
Task #2: Create a restaurant menu with detailed ingredients



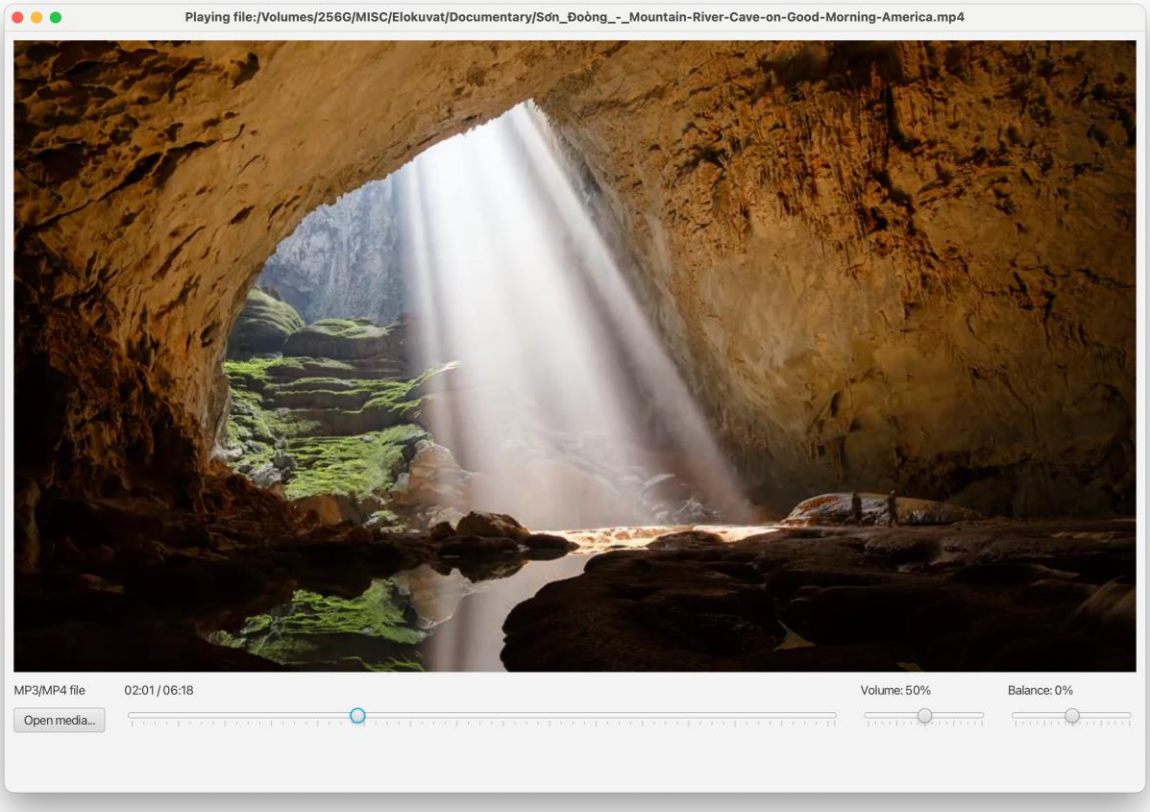
Basic rules:

- The restaurant has different cuisine style: Finnish, German, Chinese, Japanese. Each cuisine has special dishes. Each dish has certain ingredients.
- User select cuisine style then dishes, the program will collect ALL ingredients in ALL selected dishes and show in the right text area





Task #3: I created a compact media player



Basic functionalities:

- Select a movie file by opening a dialog. Support MP3 and MP4 file types. *The sample movie is about Son Doong, the world largest cave in Vietnam, broadcast by ABC channel in the "Good Morning, America" program*.
- Seek the media file by dragging the main slider into the desired position.
- Adjust volume with volume slider.
- Set volume balancing by the balance slider on the right.

Task #4 and Task #5: is customize code from Exercise 6 – Task #6 with no creativity

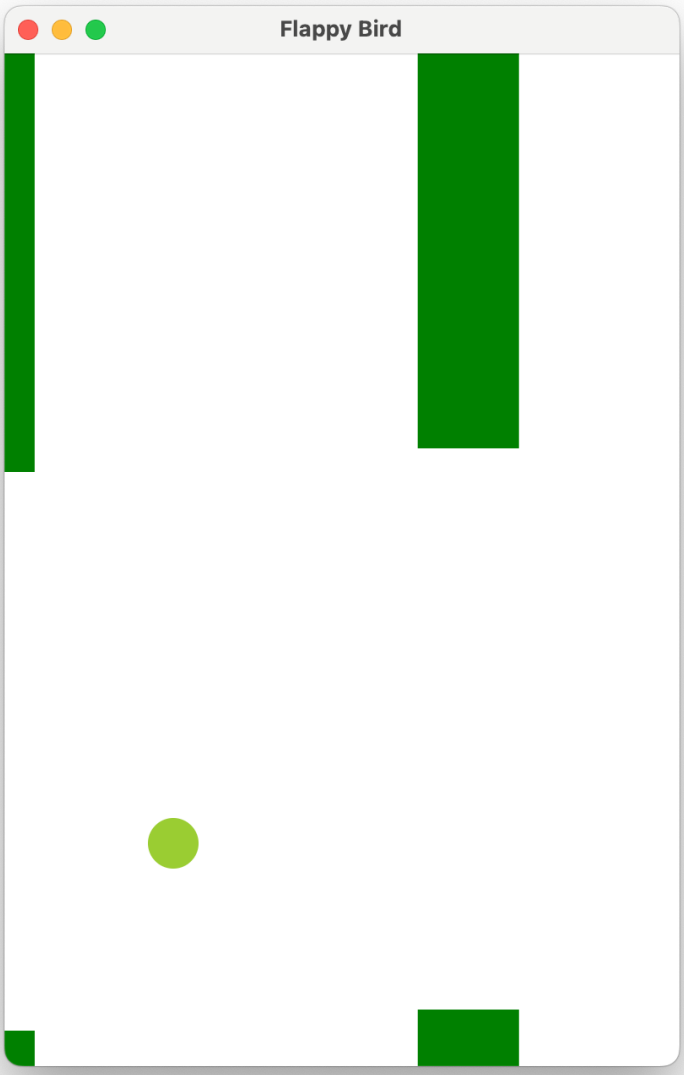
Task #6:

This task will rewrite the task 2, have some improvements:

* - Using OOP for ingredients
* - Enhanced by adding images
* - Based on idea of TasteAtlas.com

(*) This code has inspired me to improve and enhance as my Final Project, which you can see in the "**Project Brief**"

Task #7: Flappy bird remake (*Code related to the game's TimeLine is suggested by ChatGPT-4o*)



- The game engine is based on PathTransition. The Pipes appear on the top and the bottom of the screen, and they transition based on the straight lines.
- The bird, green circle, has its gravitational, managed by TimeLine object and keeps pulling the bird down. User presses the SPACE bar to fly up the bird.

- The program keeps checking collisions between the bird (green circle) against the pipes.

Exercise 8: This exercise is multi thread related, and there is no GUI.